

# Anatole Coulais

Incoming PhD Student in Machine Learning

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📍 Rennes, France

🌐 Anatole-Coulais

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## Education

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| <b>École Centrale de Lyon</b> , Mathematics                                                                                                                                                                                        | Lyon, France     |
| <b>Program</b>                                                                                                                                                                                                                     | 2025 – 2026      |
| <ul style="list-style-type: none"> <li>Stochastic calculus and stochastic differential equations, analysis and modeling of PDEs.</li> <li>Convex optimization, sparsity, statistics, deep learning, machine learning.</li> </ul>   |                  |
| <b>Université Claude Bernard Lyon 1</b> , Mathematics                                                                                                                                                                              | Lyon, France     |
| <b>Completed independently, in parallel with em-lyon)</b>                                                                                                                                                                          | 2022 – 2024      |
| <ul style="list-style-type: none"> <li>Probability, measure theory, topology, introductory functional analysis, differential calculus.</li> <li>Linear algebra, differential equations, group theory, rings and fields.</li> </ul> |                  |
| <b>Grand Lyon business school</b> , Data Science                                                                                                                                                                                   | Lyon, France     |
| <b>École Pro-gram</b>                                                                                                                                                                                                              | 2022 – 2026      |
| <ul style="list-style-type: none"> <li>Statistics, artificial intelligence, data science, economics.</li> </ul>                                                                                                                    |                  |
| <b>Classe Lycée Michel Montaigne</b> , Economics and Business (ECG)                                                                                                                                                                | Bordeaux, France |
| <b>Pré-para-toire</b>                                                                                                                                                                                                              | 2020 – 2022      |
| <ul style="list-style-type: none"> <li>Mathematics, economics, philosophy, literature.</li> </ul>                                                                                                                                  |                  |

## Experience

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| <b>CNRS / IRISA</b> , PhD Candidate (incoming)                                                                                                                                                                                                                                                                                             | Rennes, France                |
| PhD supervised by Nicolas Keriven (CNRS) and Titouan Vayer (Inria), on invariance and transferability in generative models for structured data such as graphs and point clouds.                                                                                                                                                            | Starting October 2026         |
| <b>IRISA / Inria, COMPACT team</b> , Research Intern                                                                                                                                                                                                                                                                                       | Rennes, France                |
| Six-month research internship supervised by Nicolas Keriven and Titouan Vayer, on diffusion on graphs and the generalization of generative models for graph machine learning.                                                                                                                                                              | Apr 2026 – Sept 2026          |
| <ul style="list-style-type: none"> <li>Invariant stochastic differential equations under group actions.</li> <li>Equivariant flow matching on sets and graphs, using exchangeability and de Finetti-type structure.</li> <li>Size generalization for exchangeable generative models via Wasserstein-over-Wasserstein distances.</li> </ul> | 7 months                      |
| <b>Schneider Electric R&amp;D</b> , Data Scientist Intern                                                                                                                                                                                                                                                                                  | Paris, France / Madrid, Spain |
| <ul style="list-style-type: none"> <li>Co-authored an article on machine learning models of greenhouse gas emissions.</li> <li>Modeled organic carbon sequestration in agricultural soils.</li> <li>Co-supervised a student capstone project in data science with Institut Polytechnique de Paris.</li> </ul>                              | Oct 2024 – Aug 2025           |
|                                                                                                                                                                                                                                                                                                                                            | 1 year                        |

**Université de Montréal, Department of Mathematics and Statistics**, Research Intern

Supervised by Florian Maire.

Montréal, Canada

Aug 2023 – Dec 2023

6 months

- Studied "Unbiased Markov Chain Monte Carlo Methods With Couplings" and implemented the methods in Python.
- Presented the paper to about 20 PhD students and researchers in the department.

## Skills

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**Programming:** Python (NumPy, pandas, scikit-learn, PyTorch), LaTeX, Git

**Machine Learning:** Generative models, graph neural networks, equivariant and invariant architectures, diffusion and flow matching models

**Mathematics:** Stochastic calculus, optimal transport, measure theory, group representation theory

## Languages

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**French:** Native speaker

**English:** C1

**Spanish:** B1

## Interests

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**Research:** Generative models, graph machine learning, stochastic processes